EXP:4 - INTEGER PROGRAMMING PROBLEM- BRANCH AND BOUND METHOD

AIM:

To solve any integer programming problem using the Branch and Bound method in Python.

PROCEDURE:

1. Formulate the integer programming problem.
2. Implement the Branch and Bound method in Python.
3. Solve the problem using the implemented method.
4. Verify the solution.

Code:

```
import numpy as np

from scipy.optimize import linprog

from queue import Queue


# Solve LP relaxation

def solve_lp(c, A, b, bounds):

    return linprog(c, A_ub=A, b_ub=b, bounds=bounds, method='highs')


# Branch and Bound

def branch_and_bound(c, A, b, bounds):

    Q = Queue()

    Q.put((c, A, b, bounds))


    best_solution = None

    best_value = float('-inf')


    while not Q.empty():

        current_problem = Q.get()

        res = solve_lp(*current_problem)


        if res.success and -res.fun > best_value:

            solution = res.x
```

```
        if all(np.isclose(solution, np.round(solution))):
            value = -res.fun
            if value > best_value:
                best_value = value
                best_solution = solution
        else:
            for i in range(len(solution)):
                if not np.isclose(solution[i], np.round(solution[i])):
                    lower_bound = current_problem[3].copy()
                    upper_bound = current_problem[3].copy()

                    lower_bound[i] = (lower_bound[i][0], np.floor(solution[i]))
                    upper_bound[i] = (np.ceil(solution[i]), upper_bound[i][1])

                    Q.put((c, A, b, lower_bound))
                    Q.put((c, A, b, upper_bound))
                    break

    return best_solution, best_value

# New problem data
c = [-5, -4]
A = [[3, 2], [1, 2]]
b = [18, 10]
bounds = [(0, None), (0, None)]

# Solve
solution, value = branch_and_bound(c, A, b, bounds)

print("Optimal solution:", solution)
```

```
print("Optimal value:", value)
```

Output:

Optimal solution: [4. 3.]

Optimal value: 32.0

Result:

The given integer programming problem was solved successfully using Python. The optimal solution obtained is x = 4, y = 3 with a maximum value of 32.